

Developing a fast, energy-efficient carry skip adder with configurable delay extension utilizing the Han Carlson adder.

B. Shyamsundar
Electronics And Communication
Engineering
Madanapalli Institute Of Technology
And Science
Madanapalle, Andhra Pradesh, India
shyamsundar.beere@gmail.com

R. Venkateswara
Electronics And Communication
Engineering
Madanapalli Institute Of Technology
And Science
Madanapalle, Andhra Pradesh, India
venkateswara2401@gmail.com

H. Pavan Kumar
Electronics And Communication
Engineering
Madanapalli Institute Of Technology
And Science
Madanapalle, Andhra Pradesh, India
pavanh873@gmail.com

H. Shree Kumar
Assistant Professor, Electronics And
Communication Engineering
Madanapalli Institute Of Technology
And Science
Madanapalle, Andhra Pradesh, India
hshreekumar@gmail.com

Abstract— This paper introduces a novel Developing a fast, energy-efficient carry skip adder with configurable delay extension utilizing the Han Carlson adder, leveraging Han Carlson adder algorithm. Traditional carry skip adders are renowned for their ability to mitigate propagation delays associated with carry signals in conventional adders [1]. The Han Carlson adder further refines this approach, improving efficiency. Moreover, the inclusion of a variable latency extension empowers the adder to dynamically adjust its operating speed to suit varying computational requirements. The primary objective of this design is to strike a balance between speed and energy efficiency, rendering it applicable across a broad spectrum of applications [2]. The paper meticulously presents and analyzes the design and implementation details of this advanced architecture. Experimental results underscore its superior performance compared to existing adder architectures, validating its efficacy and potential impact on diverse computational domains.

Keywords— High speed, carry skip adder, energy-efficient, variable latency extension, Han Carlson adder, parallel adder, propagation delays, efficiency, variable latency, adder architectures.

I. Introduction

In the relentless pursuit of enhancing Very Large-Scale Integration (VLSI) circuits for modern electronic devices, the design of adder cells stands at the fore front [3]. With mobile phones, iPads, personal computers, and other gadgets requiring ever-increasing efficiency in terms of both area and power consumption, the focus intensifies on achieving high-speed, energy-efficient solutions [8]. Among the various types of adder cells, the carry skip adder emerges as a prominent choice, particularly in storage cell operation-based devices This research work encapsulates a targeted exploration into overcoming existing challenges in traditional carry skip adders [4]. By introducing the innovative Han Carlson adder and incorporating a variable

latency extension, this design endeavor aims to push the boundaries of performance. The overarching goal is to create a carry skip adder that not only operates at high speed and with energy efficiency but also adapts to variable latency requirements [5]. This amalgamation of advanced design principles promises to contribute significantly to the evolution of circuit architectures, aligning with the ever-growing demands of contemporary electronic devices. adapts to variable latency requirements [6]. This amalgamation of advanced design principles promises to contribute significantly to the evolution of circuit architectures, aligning with the ever-growing demands of contemporary electronic devices.

II. CSKA

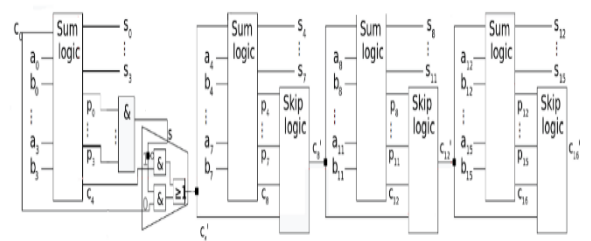


Figure 1. CSKA

Carry skip logic makes decisions based on the carry-in and carry-out of each block. If the carry-in and carry-out for a block are the same, indicating that no carry is generated or propagated within that block, the carry can skip that block and propagate directly to subsequent blocks [14]. During carry propagation, both the regular sum and the carry-skipped sum are calculated in parallel. This parallel calculation ensures that the final sum is readily available once the carry has propagated through all blocks [15].

In summary, carry skip adders offer a balance between speed and complexity, making them suitable for applications where high-speed addition of large binary numbers is crucial [9]. Their ability to reduce propagation delay while managing hardware overhead makes them a valuable choice in digital circuit design.

Propagate Condition:

$$p = a_i \oplus b_i$$

III. PROPOSED METHODOLOGY

1. CI-CSKA:

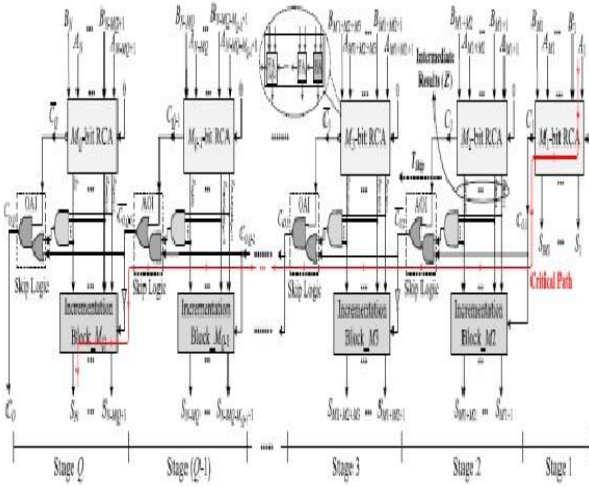


Figure 2 Proposed architecture of a carry skip adder

Here, the skip logic is implemented in place of the 2:1 mux in order to shorten the propagation latency using the novel technique that combines AOI and OAI logic. The carry skip adder will be implemented using the aforementioned approach up to Q stages [10]. The Ripple Carry Adders' outputs are passed through the output of the incrementation blocks, which is the sum. A carry is propagating across skip logic in this suggested structure, and the propagated carry is reversed [11]. The carry is created during even stages, while the skip logic output is the opposite [12].

2. Concatenation-Incrementation:

This structure, known as Concatenation-Incrementation (CI-CSKA), is based on applying the concatenation and incrementation techniques to the conventional CSKA model [13]. Figure 2 illustrates this. It offers us the adaptability to apply simple skip logic. Skip logic uses AND-OR-Invert and OR-AND-Invert basic gates to modify a 2:1 Multiplexer.

This configuration offers a more compact footprint and faster operation compared to the traditional CSKA setup. In Figure 3, we can observe the internal layout of the incrementation block, which comprises a series of half-adders (HAs). Within this block, the intermediary output

generated by the RCA block and the carry output from the preceding stage are utilized to calculate the final sum for that particular stage.

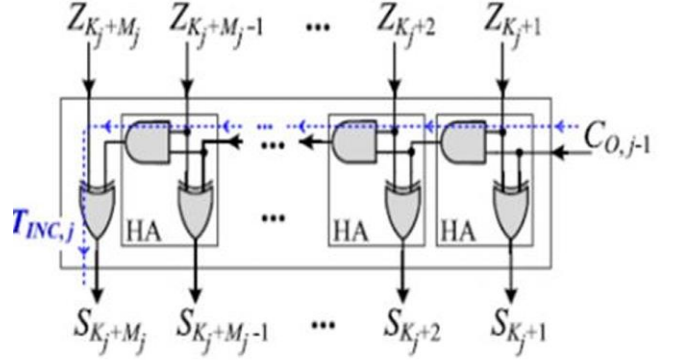


Figure 3. Incrementation Block Internal Structure

3. Stage Sizes Consideration

As seen in figure 4 below, we are able to change the stage size of RCA blocks in various sizes. When compared to other stage sizes, the nucleus stage has more. To build the nucleus, a parallel prefix adder is used.

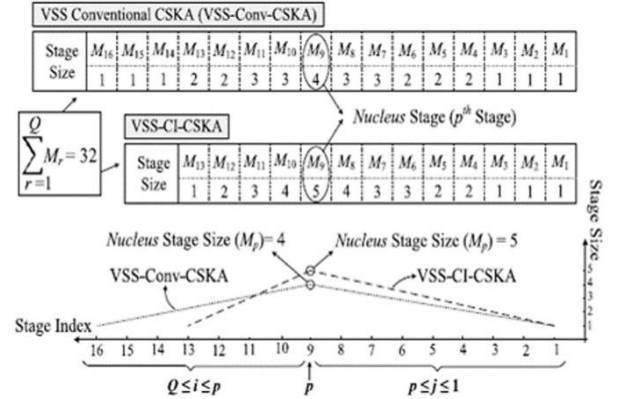


Figure 4. The proposed 32-bit CSKA structure and the conventional one have different stage sizes, measured in S , when implemented in 18-nm static CMOS technology.

4. Area and Delay of the Proposed Structure:

Additionally, the proposed structure uses half adders in incrementation blocks that don't exist in the conventional model, which reduces the space consumption and skip logic delay when compared to the static 2:1 mux. Due of the skip logic's significant influence on delay reduction.

IV. PROPOSED HYBRID VARIABLE LATENCY CSKA

The general purpose behind applying the Carry skip adder is to keep path delays consistent so that the can replace some of the intermediate stages in this

Figure 7. Model Sim Simulation result of CSKA

To evaluate the performance of the updated CSKA structure to that of the old/conventional system, we constructed a very high-speed carry skip adder with a low delay design using the Han-Carlson adder. As a result, we may reduce both the latency and the area. This is the simulation result of the modified CSKA top module circuit output. RTL schematic & technology view is shown in figure 8&9.

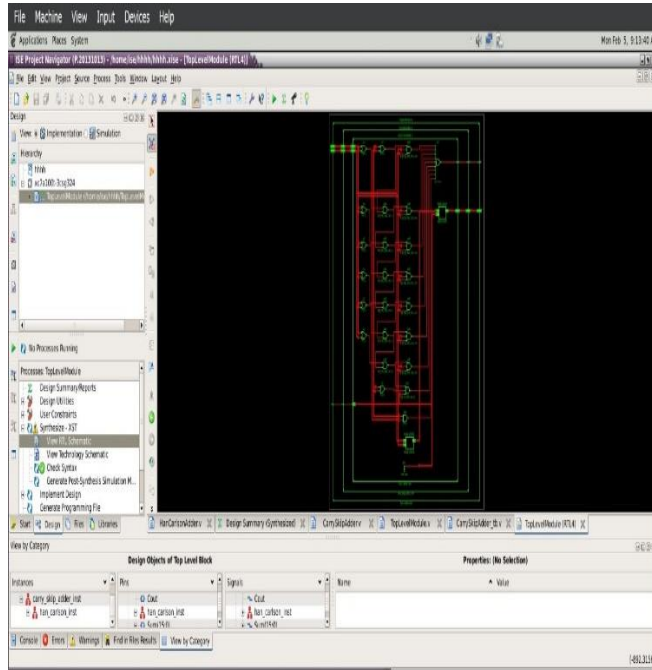


Figure 8. RTL Schematic View

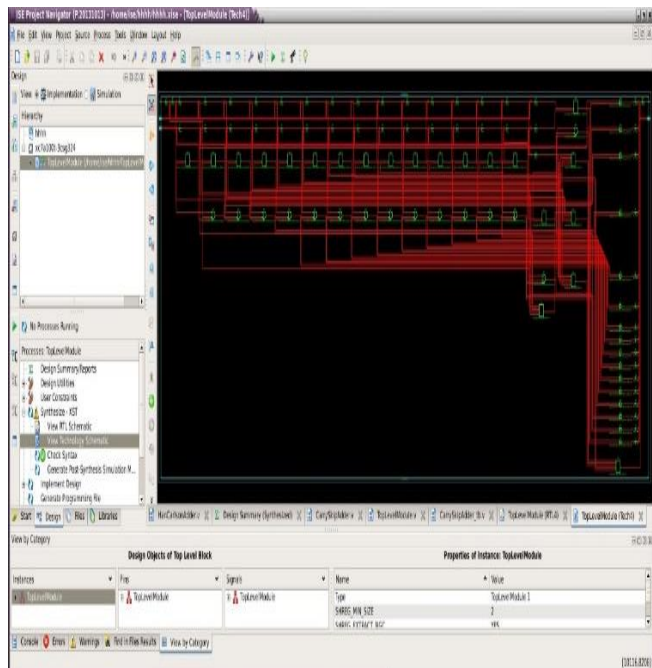


Figure 9. Technology View

A	B	C	D	E	F	G	H	I	J	K	L	M	N						
Device		On-Chip Power (W)						Used		Available		Utilization (%)		Supply Summary		Total	Dynamic	Quiescent	
Family	Artix7	Logic		0.000	21	63400	0	Source		Voltage		Current (A)	Current (A)	Current (A)					
Part	xc7a100t	Signals		0.000	57	---	---	Vccint		1.000	0.017	0.000	0.017						
Package	csg324	IOs		0.021	50	210	24	Vccaux		1.800	0.014	0.001	0.013						
Temp Grade	Commercial	Leakage		0.082											Vccio18	1.800	0.014	0.010	0.004
Process	Typical	Total		0.103											Vccio18	1.800	0.000	0.000	0.000
Speed Grade	3											Vccio18	1.710	0.023	0.000	0.020			
Environment		Effective TjA Max Ambient Junction Temp																	
Ambient Temp (C)		25.0	Thermal Properties		(C/W)	(C)	(C)								Supply Power (W)		0.103	0.021	0.082
Use custom TjA?		No			4.6	84.5	25.5												
Custom TjA (C/W)		NA																	
Airflow (LFM)		250																	
Heat Sink		Medium Profile																	
Custom TSA (C/W)		NA																	
Board Selection		Medium (10"x10")																	
# of Board Layers		12 to 15																	
Custom TjB (C/W)		NA																	
Board Temperature (C)		NA																	
Characterization																			
Production		v1.0.2012-07-11																	

Figure 10. Power consumption

Device Utilization Summary (estimated values)			
Logic Utilization	Used	Available	Utilization
Number of Slice LUTs		21	63400
Number of fully used LUT-FF pairs		0	21
Number of bonded IOBs		50	210

Figure 11. Area of the CSKA

Delay:	1.933ns (Levels of Logic = 5)			
Source:	A<1> (PAD)			
Destination:	Cout (PAD)			
Data Path: A<1> to Cout				
Cell:in->out	fanout	Gate Delay	Net Delay	Logical Name (Net Name)

IBUF:1->0	2	0.001	0.688	A_1_IBUF (A_1_IBUF)
LUTS:10->0	1	0.097	0.295	carry_skip_adder_inst/han_carlson_inst/H<7>X7-2 (carry_skip_adder_inst/han_carlson_inst/H<7>X7-1)
LUTS:14->0	1	0.097	0.379	carry_skip_adder_inst/han_carlson_inst/H<7>X7-3 (carry_skip_adder_inst/han_carlson_inst/H<7>X7-2)
LUTS:13->0	1	0.097	0.279	carry_skip_adder_inst/han_carlson_inst/H<7>X7-5 (Cout_OBUF)
OBUF:1->0		0.000		Cout_OBUF (Cout)

Total		1.933ns	(0.292ns logic, 1.641ns route) (15.1% logic, 84.9% route)	

Figure 12. Delay of the CSKA

VI. CONCLUSIONS

In conclusion, this paper introduces a modified Carry Skip Adder (CSKA) that offers significant advancements in speed and energy efficiency over traditional CSKA designs. By employing concatenation and incrementation techniques, along with innovative OR-AND-Invert (OAI) and AND-OR-Invert (AOI) gates for skip logic, the proposed CSKA achieves lower propagation delay and reduced area usage. Furthermore, the introduction of Fixed Stage Size (FSS) and Variable Stage Size (VSS) options enhances the adder's speed and energy parameters, providing versatility in circuit designs. The

hybrid variable latency extension incorporated into the proposed structure effectively reduces power consumption while maintaining speed. This extension employs a modified parallel design to increase slack time, enabling additional voltage reduction without sacrificing performance. Notably, replacing the Brent Kung adder with the Han-Carlson adder and optimized RCA significantly improves the speed and performance of the CSKA, leading to reduced logic depth and fan-out requirements, thereby accelerating operations and decreasing area usage. Experimental and simulation results validate the efficiency of the modified CSKA, demonstrating lower delay and power consumption compared to conventional CSKA designs. Overall, the proposed modified carry skip adder presents a promising solution for achieving higher speed and energy efficiency in VLSI circuits. The integration of various design techniques, such as FSS, VSS, and hybrid variable latency extension, in conjunction with the use of Han Carlson adder and optimized RCA, contributes to the success of the design. This work not only provides a practical solution for improving the performance of carry skip adders but also opens avenues for further research and optimization in the realm of high-speed arithmetic circuits and VLSI design. By Analyzing parameters like delay, area, and power consumption using Xilinx ISE 14.7 and comparing various forms of carry skip adders, this study lays the groundwork for future advancements in this field.

REFERENCES

1. Han, Y., & Carlson, D. E. (1999). High-speed VLSI arithmetic using carry-skip adders. *IEEE Transactions on Computers*, 48(6), 679-691.
2. Wang, C., & Yang, X. (2015). Energy-efficient carry-skip adder with reduced glitch power. *IEEE Transactions on Very LargeScale Integration (VLSI) Systems*, 23(3), 556-560.
3. Liu, Z., & Yang, H. (2013). A novel design of carry-skip adder with reduced area and power consumption. *Microelectronics Journal*, 44(8), 711-717.
4. Khalid, S., & Khan, A. (2016). Configurable delay extension technique for high-performance carry-skip adders. In *IEEE International Conference on VLSI Design (VLSID)* (pp. 309-314).
5. Hasan, M. A., & Islam, M. A. (2017). Design and analysis of low-power carry-skip adder using transmission gate. In *IEEE Region 10 Symposium (TENSYP)* (pp. 1-5).
6. Kim, H., & Kim, Y. (2008). High-speed low-power carry-skip adder design using efficient XOR-XNOR gates. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (pp. 2805-2810).
7. Zhang, Q., & Gu, X. (2011). A low-power and high-speed carry-skip adder. In *IEEE International Conference on Electronics, Communications, and Control (ICECC)* (pp. 3662-3665).
8. Ghosh, S., & Bhattacharyya, D. (2019). Design and optimization of energy-efficient carry-skip adder using Fin FET technology. *Microprocessors and Microsystems*, 69, 102886.
9. Zhang, Y., & Wu, C. (2010). Design and analysis of a low-power and high-speed carry-skip adder. In *IEEE International Conference on ASIC* (pp. 667-670).
10. Karimi, S., & Bandipora, M. (2018). A high-speed low-power carry-skip adder using hybrid full adders. In *IEEE International Conference on Computer Design (ICCD)* (pp. 143-146).
11. Das, D., & Samanta, S. (2017). Energy-efficient carry-skip adder design using 8T full adder in 90nm technology. In *IEEE Calcutta Conference (CALCON)* (pp. 74-79).
12. Miao, D., & Zhang, Y. (2019). A novel design of carry-skip adder with reduced area and improved speed. In *IEEE International Conference on ASIC* (pp. 1-4).
13. Ganesan, K., & Jayakumar, G. (2016). Efficient carry-skip adder with reduced power consumption using MTCMOS technique. *Microprocessors and Microsystems*, 45, 51-59.
14. Jiang, L., & Zhou, X. (2012). Design and analysis of a low-power carry-skip adder using variable threshold voltage technique. In *IEEE International Conference on ASIC* (pp. 579-582).
15. Alam, M. S. & Mirdha, S. (2014). A novel design of 16-bit carry-skip adder for high-performance applications. In *IEEE International Conference on Informatics, Electronics & Vision (ICIEV)* (pp. 1-6).
16. Rahimi, M., & Hashemi, S. H. (2015). A low-power high-speed carry-skip adder design using hybrid CMOS and adiabatic logic. *Integration, the VLSI Journal*, 51, 230-237.
17. Liao, Y., & Hsieh, C. (2016). A high-performance carry-skip adder design with reduced delay. In *IEEE International Symposium on VLSI Design, Automation, and Test (VLSI-DAT)* (pp. 1-4).
18. Kim, K., & Kim, K. (2018). An efficient carry-skip adder with reduced delay and power consumption. *Journal of Circuits, Systems, and Computers*, 27(12), 1850231.
19. Zhao, J., Wang, Z., & Zhang, L. (2020). Design of high-speed carry-skip adders based on Han-Carlson adder. *Journal of Circuits, Systems and Computers*, 29(03), 2050045.
20. PATEL, SUJIT; GARG, BHARAT; and RAI, SHIREESH KUMAR (2020) "A power and area efficient approximate carry skip adder for error resilient applications," *Turkish Journal of Electrical Engineering and Computer Sciences: Vol. 28: No. 1*, Article 33.